A Large-Language-Model-Enabled Robotic System with Application to Power Cable Manipulation

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2026 16th International Conference on Power, Energy, and Electrical Engineering (CPEEE) | 979-8-3315-6119-2/26/$31.00 ©2026 IEEE | DOI: 10.1109/CPEEE69412.2026.11521580

***Abstract*—Industrial automation is evolving toward greater intelligence and collaboration, especially in sectors such as energy and power where reliability is critical. However, most robotic systems still rely on rigid task scripts and limited perception, restricting their adaptability and interaction with humans. This paper introduces a generalized framework for human-robot interaction that integrates large vision-language models (VLMs) with robotic control. The system combines a GPT-4o-based Communication Agent, which interprets user intentions and generates structured commands, with a Physical Agent that executes tasks using modular motion skills. A CNN-based vision module (YOLOv11x-OBB) with depth sensing provides 3D perception for precise manipulation. The framework is demonstrated through a collaborative cable insertion task, showing that natural language instructions can be effectively translated into adaptive robot actions. Results suggest that LLM/VLM-driven agents can significantly enhance flexibility in industrial robotic automation.**

Keywords—power cable manipulation, LLM-based automation, robot manipulation, robot perception

I. Introduction

Industrial automation is now a key part of how the energy and power sector improves its operations. In systems where reliability and efficiency are critical, automation helps keep high-value infrastructure running smoothly and without interruption. The move toward digital technologies, often called Industry 4.0, has introduced tools such as intelligent sensors, digital twins, and cloud-based control systems. These technologies have greatly improved how power systems are monitored and maintained through automation [1]-[3].

In this context, robotic manipulators are becoming important for complex tasks such as inspecting and handling infrastructure. Reviews of mobile manipulators show their growing role as a core part of the Industry 4.0 framework, combining mobility, precise manipulation, advanced sensing, and network connectivity to bring flexible automation into changing environments [4]. Studies have shown that serial and hybrid manipulator arms are now used in factories, logistics centers, and utility sites [5]. In power-system and infrastructure applications, robotic arms mounted on mobile bases or gantries have been used for component replacement, extending automation from simple sensing to direct physical work with electrical equipment [6]. Looking ahead, industrial automation is expected to advance to the next level, where intelligent robots can follow engineers, understand natural language commands, and perform tasks with greater flexibility.

Although robotic systems have made significant progress in automation, most current deployments still rely on predefined task scripts, teach-pendant programming, and specialized perception models designed for narrow operating conditions. Such rigid setups limit a robot's ability to adjust goals, re-plan tasks, or work flexibly with human operators in complex and changing environments such as substations and industrial plants. Reviews of human-robot collaboration [7], [8] identify several ongoing challenges, including complex setup procedures, limited autonomy near humans, and poor adaptability when the working context changes unexpectedly. These systems also struggle to interpret multi-step instructions or perform context-aware cooperation. As a result, current robotic frameworks allow coexistence with humans but not true collaboration, highlighting the need for methods that incorporate semantic understanding and natural human-robot interaction for more intuitive teamwork.

Recent advances in large language models (LLMs) and vision-language models (VLMs) have created new opportunities to bring higher-level reasoning and natural communication into industrial automation. These models show strong capabilities in understanding instructions, reasoning over context, and generating coherent plans in response to complex, open-ended prompts [9]–[11]. Trained on large amounts of multimodal data, they capture not only language patterns but also general knowledge about the physical and operational world. This allows human operators to interact with robotic systems using intuitive, language-based interfaces instead of low-level scripts or custom control software.

Building on VLMs, a recent trend in imitation learning and behavior cloning, led by several major technology companies, is the development of Vision-Language-Action (VLA) models [12]–[14]. These models aim to use pre-trained VLMs to understand semantics and directly predict the robot's next action. Although flexible to various human instructions, training these models requires extensive computational resources and time. Even with improved datasets and training methods, which enhance generalization, these models still need fine-tuning for new and complex tasks. In practice, they generalize well to tasks seen during training but remain difficult to scale to unseen or more diverse scenarios.

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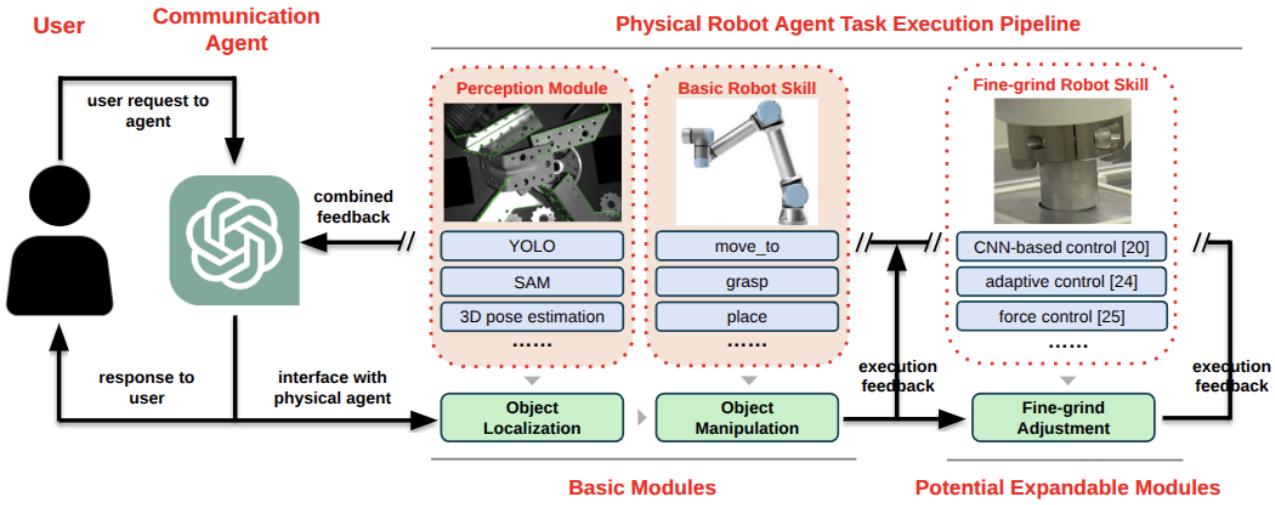


Fig. 1. An LLM-enabled robotic manipulation system

To this end, this paper presents a generalized framework for human-robot interaction in industrial automation, demonstrated through a representative collaborative cable insertion task. The proposed framework leverages the reasoning and contextual understanding capabilities of LLMs to interpret human instructions, extract key task information from dialogue, and translate it into executable robot commands. In this setup, the robot's vision module uses a convolutional neural network (CNN)-based model [15], [16] fine-tuned to detect and locate relevant objects in the workspace. Based on the perception input, the robot agent selects and executes appropriate manipulation skills from a predefined skill library to interact with the target object. The robot's execution feedback is then communicated back to the language model, enabling either task re-planning or continued interaction with the human operator.

This modular design allows the robot to complete complex tasks by decomposing them into multiple skills it already possesses. Moreover, the modularity provides flexibility beyond what a single VLA model can achieve — treating all possible solutions, whether rule-based path planning, 3D-vision based detection, or learning-based models such as VLAs and diffusion policies, as interchangeable building blocks. This enables the system to upgrade or replace individual perception modules, add new manipulation skills in a plug-and-play manner, and even integrate multiple robot agents into the interaction process.

This feature is especially valuable for cable insertion in industrial automation. In real deployments, the vision modules, target cables, and even the robot itself may change over time due to shifting production demands. To maintain accuracy under diverse environments and disturbances, improved vision or motion planning methods are often required. The proposed framework offers not only a convenient user interface but also the flexibility needed to adapt to evolving cable-insertion requirements.

The remainder of this paper is organized as follows. Section II presents the conceptual design and methodology of the framework. Section III describes each module in detail along with the experimental implementation. Section IV demonstrates the application of the framework to a collaborative cable insertion task, analyzes the results, and discusses the current limitations. Finally, Section V concludes the paper.

II. The Methodology

This section introduces the conceptual design of the framework, while the detailed implementation and tools used are presented in Section III. A snapshot of the framework, showing the communication components, robot execution modules, and overall information flow, is provided in Fig. 1. The framework consists of two agents: the LLM-enabled Communication Agent (CA) and the Physical Robot Agent (PA). These abbreviations will be used throughout the remainder of this paper. Keep your text and graphic files separate until after the text has been formatted and styled. Do not use hard tabs, and limit use of hard returns to only one return at the end of a paragraph. Do not add any kind of pagination anywhere in the paper. Do not number text heads-the template will do that for you.

A. Communication Agent (CA)

The Communication Agent (CA) focuses on interpreting user requests and reasoning about the tasks that need to be executed to fulfill those requirements. At the beginning of the interaction, the CA guides the user toward specifying a concrete action, such as grasping an object requested by the user. Once the CA confirms the user's intention regarding the exact action, the object to be manipulated, or the target location for the Physical Agent (PA), it generates an explicit keyword to inform and trigger the PA for task execution. The corresponding task parameters, such as the desired skill and object, are then passed to the PA as variables through agent communication.

Beyond its assistive role, the CA can also provide guidance and explanations to the user through search and image generation capabilities. Since LLMs/VLMs are pre-trained on vast and diverse datasets, such guidance and clarification can be invaluable in preventing users from making critical mistakes, particularly when they are unfamiliar with the task or its detailed procedures.

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B. Physical Robot Agent (PA)

The Physical Robot Agent (PA) is responsible for monitoring the conversation, detecting trigger keywords generated by the CA, and executing the corresponding tasks by invoking its pre-implemented skill modules. These skill modules are designed to be flexible, allowing dynamic passing of variables such as object identifiers, target locations, and other task specific parameters. They operate in a plug-and-play manner, meaning any module can be updated, replaced, or improved with minimal effort, as long as a unified input/output format is maintained.

Each skill call follows a structured pipeline:

1) Aperception module, typically a CNN-based model for 2D image input or a 6D object pose estimation method for 3D point clouds, is used to extract the pose of the target object.
2) The estimated poses are then passed to pre-programmed basic skill modules, selected by the CA (e.g., *move* and *grasp*), to perform an initial high-speed, collision-free movement.
3) The robot's motion can then be refined with subsequent add-on modules, potentially including a low-speed fine adjustment module that ensures the high-precision require ments of the task are met.
4) Once these steps are successfully completed—indicating that the object has been manipulated as intended—the PA proceeds to the next skill requested by the CA, returning to step (1).
When all requests have been executed, the PA reports the final execution status to the CA, enabling the CA to move on to the next user request or re-evaluate a previous task if necessary.

The combination of high-speed motion and low-speed fine adjustment is motivated by both efficiency and accuracy considerations. Although fine manipulation methods, such as CNN-based motion control or force-control-based search, can achieve high precision, they are often time-consuming. Therefore, when the robot operates in free space or in simple contact scenarios (e.g., Lego block stacking), fast motion is preferred for efficiency. However, perception methods that explicitly estimate object poses often exhibit limited sub millimeter accuracy, which is insufficient for high-precision tasks such as cable insertion. When direct execution of a pre-programmed action fails due to unexpected collisions or inaccuracies in vision-based localization, the low-speed fine-adjustment module compensates by handling contact-rich interactions to achieve the required precision.

III. Implementation for Calble Insertion

A. LLM as the Communication Agent

Since GPT-4o supports multimodal inputs, including both text and images, it was used as the CA to build an interface that enables customized interaction patterns. A system prompt was designed to define the agent's role and the specific scenario it should handle. The exact system prompt is omitted here, but it generally instructs the LLM to:

1) Generate a keyword to trigger the PA following a predefined format whenever the user's intention is clear.

Algorithm 1: Framework Algorithm Implementation

```
Input: System prompt file; keyword list file; perception
       module; basic robot skill; fine-grained robot skill
Output: None
dialogue ← ∅
systemPrompt ← ReadSystemPromptFile()
append systemPrompt to dialogue
PAKeywords ← ReadKeywordListFile()
while conversation ongoing do
    userPrompt ← GetUserInput()
    append userPrompt to dialogue
    response ← GPT4oResponse(dialogue)
    display(response)
    append response to dialogue
    if response contains a PA keyword then
        numSkills ← CountKeywords(response)
        for n ←1 to numSkills do
            object ← ExtractObject(response, n)
            skill ← ExtractSkill(response, n)
            pose ← PerceptionModule(object)
            feedback ← RobotExecute(skill, pose)
            if fine-grained skill available then
                feedback ←
                  RobotExecute(fineGrainedSkill, pose)
            end
            append feedback to dialogue
            display(feedback)
        end
        taskEval ← GPT4oResponse(dialogue)
        display(taskEval)
    end
end
```

2) Evaluate the task execution status returned from the PA to determine whether the task should be repeated or proceed to the next one.
3) Continue communicating with the user by elaborating on their questions, as well as performing searches or generating images when necessary to provide additional clarity.
Apart from the system prompt, the detailed algorithm pseudocode that describes Fig. 1 can be seen from Algorithm 1.

B. CNN-based Object Detection

In principle, several mainstream architectures could be used. Two-stage instance segmentation methods such as Mask R-CNN [17] provide high-quality object masks in addition to detection. Foundation segmentation models such as the Segment Anything Model (SAM) [18] further extend this idea by using a promptable Vision Transformer trained on a very large corpus of images and masks, enabling zero shot generalization to novel scenes and categories. In parallel, transformer-based detectors such as DETR [19] formulate detection as a set-prediction problem, simplifying the pipeline by removing hand-crafted anchor priors and non-maximum suppression while achieving strong accuracy on natural image benchmarks.

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(a) (b) (c) (d)

Fig. 2. (a) The original image from RealSense with objects indicated. (b) The YOLO-estimated bounding box generated on the detected object with their estimated coordinate printed. 'C', 'H', and 'U' stand for Type-C, HDMI, and USB3. (c) The extended pixel ring used to estimate socket distance from the camera (background faded). (d) The fitted relationship (almost linear) between the real-world-to-pixel ratio and object's distance from the camera.

However, for the proposed cable insertion task, these alternatives present practical limitations. Mask R-CNN and SAM produce dense segmentation masks that are computationally costly but unnecessary for pose estimation and controller [20] that relies solely on bounding box position and orientation, leading to increased latency without functional benefit. Likewise, DETR-based architecture, while elegant and competitive in accuracy, generally requires heavier training and inference budgets, and oriented box variants remain less mature for real-time robotic integration.

In contrast, YOLO (You Only Look Once) [21], [22] is a widely adopted real-time object detection framework that offers an effective balance between speed and accuracy, making it particularly suitable for applications requiring instant visual perception. A YOLO-based oriented detector provides a direct, low-latency mapping from image to object bounding box information, which makes it more compatible with rapid decision-making and closed-loop control, where short inference time and fast pose updates are essential. For this reason, YOLOv11x-OBB provides the most efficient and task-aligned solution among mainstream detection frameworks.

In our implementation, we used the pre-trained YOLOv11x OBB model as the backbone, which was trained on the DOTA dataset [23]. This model can detect oriented or inclined bounding boxes, providing useful information for estimating object orientation in real-world settings. We collected 30 images of USB, Type-C, and HDMI socket hubs from the experimental environment, as these components serve as representative examples in this study. The collected images were manually labeled and used to fine-tune the pre-trained model. An example of the predicted bounding boxes and their corresponding labels is shown in Fig. 2b.

However, despite the robustness of YOLO, its bounding box estimations do not contain 3D spatial information. To address this limitation, we employed an Intel RealSense camera, which provides both RGB and depth data, to compute the 3D coordinates of the detected bounding boxes in the camera frame. Considering the typical setup of insertion tasks, where multiple socket holes are arranged on the same flat surface of a hub, the implementation proceeds as follows: (a) the fine-tuned model detects the bounding box of the target object (e.g., a USB socket hole); (b) The bounding box is expanded by a few pixels to form a "ring" that encloses the original box, and the corresponding "depth ring" is extracted from the depth image (Fig. 2b); (c) the average depth of the pixels within this extended region is computed to estimate the distance from the hole center to the camera. The depth ring is highlighted in Fig. 2c.

Since the bounding box size and coordinates are expressed in pixels rather than physical units, an additional calibration step is required to convert pixel distances into real-world measurements. The underlying idea is that as the object moves farther from the camera, each image pixel represents a larger real-world distance on the object. To establish this relationship, we collected data relating the real-world-to-pixel ratio to the object's distance from the camera (estimated using the afore mentioned depth estimation) and fitted a polynomial curve to capture this correlation, as shown in Fig. 2d. During inference, this curve takes in the estimated distance, corresponding to the object's Z-coordinate in the camera frame, then estimates the pixel-to-real ratio. This ratio is then used to convert pixel coordinates into real-world X and Y coordinates. The estimated coordinate of the bounding boxes can be visualized in Fig. 2b.

The "depth ring" discussed in this section can also be used to estimate the normal vector of the hub surface shared by multiple socket holes. This normal vector directly indicates the surface orientation and can assist in determining the correct insertion direction. When the relative positions of the camera and the insertion target are fixed, it is reasonable to assume that the surface is parallel to the camera plane, such that the normal vector points toward the camera. This assumption simplifies both implementation and calibration.

C. *Robot Agent and Skill Modules*

Our implementation employs a robot manipulator UR5e. As a proof-of-concept demonstration, several high-speed skill modules have been implemented in the skill bank:

1) move_to: moves the robot end-effector to a specified pose;
2) grasp: calls move to and closes the gripper;
3) hand_over: calls move to and opens the gripper;
4) insert: moves the end-effector along a desired direction at low speed while monitoring force for safety;
5) reset: opens the gripper and moves to the pre-determined home pose.

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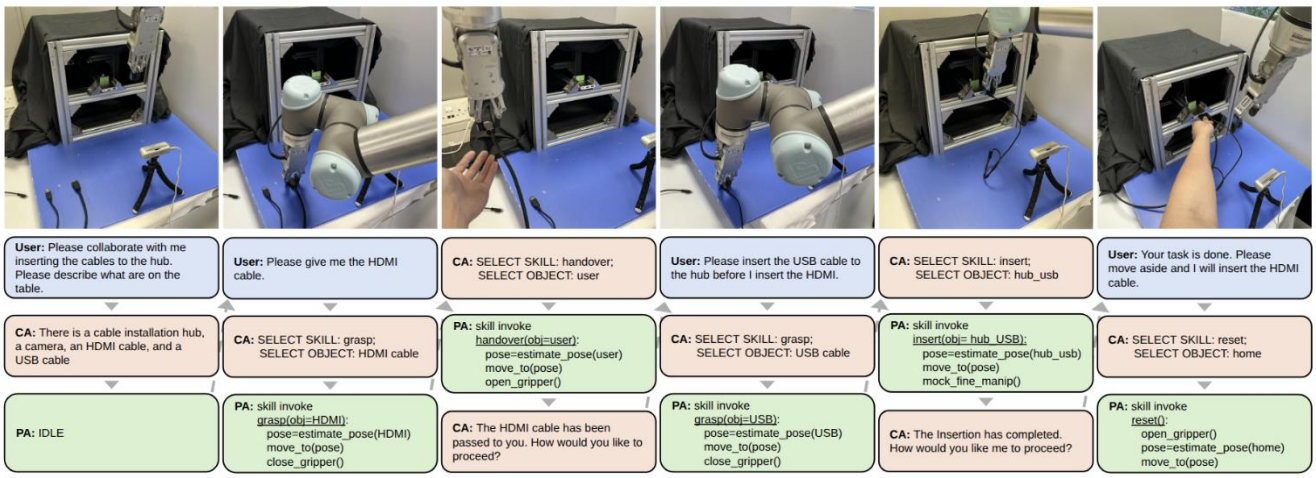


Fig. 3. Time-lapse of the collaborative cable insertion demonstration. The prompts shown below the images illustrate the conversation flow between the user and the CA, as well as how the PA interacted with CA and invoked the corresponding skill modules.

For low-speed, fine-grained motions, we adopt the CNN based control and adaptive control methods from our previous work [20], [24]. These techniques use the CNN-predicted bounding box as feedback to command the robot's motion with proven convergence. They are particularly useful when object pose estimation from the detection modules lacks sufficient precision or when the side camera is obstructed by the robot, leaving only the hand-mounted camera available. For force control, we also consider our existing motion/force control algorithms discussed in [25], [26]. These methods provide a simple yet effective approach that enables both force monitoring and the maintenance of the desired contact force, with or without dedicated force sensors. The above fine-grained control modules were developed separately and are currently being integrated into the framework; they are discussed here to highlight their importance in achieving robust and precise manipulation.

IV. Experiment

The experiment serves as a proof-of-concept demonstration of the proposed generalized framework. The time-lapse of the collaborative cable insertion process is shown in Fig. 3. The user prompts and the CA's responses are displayed in the dialogue attached to each image. The initial setup is shown in the leftmost image, where a frame holds the insertion hub, a RealSense camera is positioned in front of the hub, and two cables, USB and HDMI, are placed beside the camera. The robot base is located behind the camera.

The user first instructed the CA to hand over the HDMI cable (as shown in the second and third images). The CA recognized the user's intent and generated two skill-selection commands for the PA to grasp the cable and then hand it over. Although these commands were generated at the same time, they are shown in separate boxes within the visualization in Fig. 3 to reflect the sequential execution process, as the PA performs each command one at a time. In this demonstration, the exact coordinates of both cables and the user were fixed; however, they could also be estimated using the same YOLO based detection method if an additional camera were available.

Next, the user instructed the CA to insert the USB cable directly into the hub (as shown in the fourth and fifth images). The CA again issued two commands to the PA to grasp and then insert the cable. For the insertion skill pipeline, the pose estimation technique described in Section III.B was used to determine the pose of the USB slot, which was then used as the reference for the insertion skill. It is worth noting that robust USB insertion, commonly referred to as the peg in-hole problem [27], typically requires force control. As mentioned in Section III.C, our existing CNN-based control and force-control modules are currently under testing in this new environment and will be integrated into the pipeline. Thus, a mock function, *fine_manip*, was temporarily used in the fifth image for this demonstration. The actual insertion was carried out using the insert skill described in Section III.C, under the assumption of perfect orientation alignment and negligible position error between the estimated and actual poses.

V. Conclusion

In this paper, we proposed a generalizable framework for interactive human-robot collaboration, demonstrated through the cable insertion task. This framework utilizes LLMs as a Communication Agent for understanding user requests, task planning, and interfacing with the Physical Agent. The Physical Agent maintains its own skill bank and perception methods for flexible task execution and manipulation of various objects. Specialized perception methods were developed for the insertion task to enable single-camera pose estimation of the insertion slot.

This modular design enables the robot to tackle complex tasks by combining existing skills while remaining far more flexible than a single VLA model. By treating diverse approaches, rule-based planning, 3D vision, VLAs, or diffusion policies, as interchangeable components, the framework supports plug-and-play upgrades to perception and manipulation modules and even seamless integration of multiple robot agents.

Future work includes completing the integration of our existing CNN-based motion control and force control algorithms to significantly improve the insertion success rate [20], [25]. It would also be beneficial to enhance the model fine-tuning process and incorporate multiple 3D cameras to improve pose estimation accuracy. Furthermore, expanding the PA's skill bank, such as adding skills like shaking and pressing, would

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broaden the framework's applicability beyond industrial automation to laboratories and household environments.

ACKNOWLEDGMENT

This work was supported by the Ministry of Education, Singapore, through the Academic Research Fund Tier 1, under Project Grant RG65/22.

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